

Internal Hackathon 2026 — Problem Statement Bank

30 Real-World Problem Statements | Tier-3 College Level | SIH-Style Format

Use this document as the content source for your hackathon website. Each problem statement has a unique ID, domain tag, type (Software / Hardware / Hybrid), and difficulty — ideal for building filters (by domain, type, difficulty) on the "Choose Your Problem Statement" page.

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ID	Title	Domain	Type	Difficulty
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PS-02	Smart Irrigation & Soil Health Monitor	Agriculture	Hardware	Intermediate
PS-03	AI Symptom Checker & Teleconsultation Queue for Rural Clinics	Healthcare	Software	Intermediate
PS-04	Low-Cost Portable Vitals Monitoring Device	Healthcare	Hardware	Advanced
PS-05	Vernacular Learning Platform for Competitive Exams	Education	Software	Beginner
PS-06	Smart Classroom Attendance System	Education	Hybrid	Intermediate
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ID	Title	Domain	Type	Difficulty
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PS-27	Wearable Performance Tracker for Rural Sports Talent	Sports	Hybrid	Intermediate
PS-28	Blockchain-Based Land Record Verification	Blockchain / GovTech	Software	Advanced
PS-29	Deepfake & Regional-Language Fake News Detector	AI/ML	Software	Advanced
PS-30	Low-Cost Agri-Robot for Sowing & Weeding	Robotics	Hardware	Advanced

Domain Legend (for website filter chips)

Agriculture · Healthcare · Education · Women Safety · Disaster Management · Smart City · Transportation · Waste Management · Water Resources · Energy · Financial Inclusion · AgriTech/MSME · Cybersecurity · Assistive Tech · Tourism & Culture · Governance · Environment · Sports · Blockchain/GovTech · AI/ML · Robotics

Type Legend:  Software ·  Hardware ·  Hybrid (Hardware + Software)

Detailed Problem Statements

PS-01 | AI-Based Crop Disease & Pest Detection

Domain: Agriculture | **Type:**  Software | **Difficulty:** Intermediate

Problem Statement: Small and marginal farmers in tier-2/3 regions lose 15-25% of crop yield annually to undetected pest infestation and disease, largely because agricultural extension officers are scarce and farmers rely on guesswork or delayed advice.

Expected Solution:

- Mobile/web app where farmers upload a leaf/crop photo (even offline-first, syncing later)
- ML image classification model to identify disease/pest with confidence score
- Regional-language treatment recommendations (organic + chemical options with dosage)
- Optional integration with local Krishi Vigyan Kendra helpline/database

Suggested Tech Stack: TensorFlow Lite/PyTorch Mobile, React Native/Flutter, Firebase, a public plant-disease dataset (e.g., PlantVillage) for training.

Beneficiaries: Smallholder farmers, agricultural extension workers.

PS-02 | Smart Irrigation & Soil Health Monitor

Domain: Agriculture | **Type:**  Hardware | **Difficulty:** Intermediate

Problem Statement: Over-irrigation and under-irrigation both damage yield and waste India's scarce groundwater. Farmers lack real-time, low-cost visibility into soil moisture, pH, and nutrient levels.

Expected Solution:

- Soil moisture, temperature, and pH sensor node powered by solar/battery
- Microcontroller (ESP32/Arduino) transmitting data via LoRa/GSM to a dashboard
- Automated valve/motor trigger for irrigation based on threshold
- SMS alerts for farmers without smartphones

Suggested Tech Stack: ESP32, capacitive soil moisture sensor, DHT11/22, relay module, Blynk/ThingSpeak or a custom dashboard.

Beneficiaries: Farmers in water-stressed districts, agri-cooperatives.

PS-03 | AI Symptom Checker & Teleconsultation Queue for Rural Clinics

Domain: Healthcare | **Type:**  Software | **Difficulty:** Intermediate

Problem Statement: Primary Health Centres (PHCs) in rural India are overloaded with a single doctor serving thousands of patients, causing long queues and missed critical cases while minor ailments consume the same time slots.

Expected Solution:

- Symptom-based triage chatbot (rule-based or ML) to flag urgency level (emergency/moderate/minor)
- Digital token/queue system reducing physical waiting
- Video-consult integration for non-critical cases with nearby doctors
- Basic patient history stored securely for follow-ups

Suggested Tech Stack: Python/Node.js backend, Rasa/Dialogflow for chatbot, WebRTC for video, PostgreSQL.

Beneficiaries: Rural patients, PHC doctors and staff.

PS-04 | Low-Cost Portable Vitals Monitoring Device

Domain: Healthcare | **Type:**  Hardware | **Difficulty:** Advanced

Problem Statement: ASHA workers and rural clinics often lack affordable devices to continuously track vitals (heart rate, SpO2, temperature) for elderly or high-risk patients between hospital visits.

Expected Solution:

- Wearable/handheld device measuring pulse, SpO2, and body temperature
- Bluetooth/GSM transmission to a companion app or SMS alert for abnormal readings
- Data logging for ASHA worker/doctor review
- Battery life optimized for field use (8+ hours)

Suggested Tech Stack: MAX30100/MAX30102 pulse-oximeter sensor, ESP32, mobile app (Flutter/Kotlin), cloud sync (Firebase).

Beneficiaries: ASHA workers, elderly and chronic-illness patients in rural areas.

PS-05 | Vernacular Learning Platform for Competitive Exams

Domain: Education | **Type:**  Software | **Difficulty:** Beginner

Problem Statement: Students preparing for government exams (SSC, Railways, State PSC) in tier-3 towns often struggle with English-heavy study material, limiting access to quality preparation resources.

Expected Solution:

- Content platform offering notes, mock tests, and video content in Hindi/regional languages
- Adaptive practice tests with performance analytics
- Offline-download mode for low-connectivity areas
- Peer discussion forum for doubt-solving

Suggested Tech Stack: React/Next.js, Node.js + MongoDB, PWA for offline support.

Beneficiaries: Competitive-exam aspirants in Tier-2/3 towns.

PS-06 | Smart Classroom Attendance System

Domain: Education | **Type:** 🔄 Hybrid | **Difficulty:** Intermediate

Problem Statement: Manual attendance in large colleges consumes classroom time and is prone to proxy attendance ("buddy punching"), reducing effective teaching hours.

Expected Solution:

- Camera/RFID-based hardware unit at classroom entry for face or card recognition
- Backend dashboard for faculty to view/export attendance instantly
- Anti-spoofing checks (liveness detection) to prevent proxy attendance
- Parent/student portal showing attendance percentage in real time

Suggested Tech Stack: OpenCV/face-recognition library, Raspberry Pi/ESP32-CAM, RFID module, Django/Node.js backend.

Beneficiaries: College administration, faculty, students.

PS-07 | Women Safety Wearable with Panic Alert

Domain: Women Safety | **Type:** 🔄 Hybrid | **Difficulty:** Advanced

Problem Statement: Existing safety apps require unlocking a phone and multiple taps during an emergency, which is impractical in a real threat scenario. A discreet, instant-trigger solution is needed.

Expected Solution:

- Wearable device (band/pendant) with a single-press panic button
- GPS + GSM module to send live location via SMS/app to pre-set emergency contacts and nearest police station (simulated API for hackathon)
- Companion app showing live tracking to guardians
- Fake-call or loud-siren deterrent feature

Suggested Tech Stack: NEO-6M GPS module, SIM800L GSM module, ESP32, Flutter companion app.

Beneficiaries: Women commuters, students, working professionals.

PS-08 | Flood Prediction & Early Warning System

Domain: Disaster Management | **Type:** 🖥️ Software | **Difficulty:** Advanced

Problem Statement: Flood-prone districts often receive warnings too late because prediction relies on manual river-gauge readings rather than integrated real-time data and forecasting models.

Expected Solution:

- ML model combining rainfall data, river water-level history, and terrain data to predict flood probability by region
- Public dashboard/app with color-coded risk zones
- SMS/push alerts to residents in high-risk zones
- Integration-ready design for government IMD/CWC data feeds

Suggested Tech Stack: Python (scikit-learn/XGBoost), historical open datasets (data.gov.in), Leaflet.js/Mapbox for visualization.

Beneficiaries: District disaster management authorities, residents of flood-prone areas.

PS-09 | Landslide/Earthquake Early Warning Sensor Node

Domain: Disaster Management | **Type:** ⚙️ Hardware | **Difficulty:** Advanced

Problem Statement: Hilly regions in India face recurring landslides with minimal ground-level sensing infrastructure to give residents even a few minutes of warning.

Expected Solution:

- Sensor node measuring soil vibration, tilt, and moisture on slopes
- Threshold-based buzzer/alert triggered locally plus data transmission to a control dashboard
- Low-power design for solar/battery operation in remote areas
- Clustered node network to triangulate risk zones

Suggested Tech Stack: MPU6050 (tilt/vibration), soil moisture sensor, LoRa module, ESP32, solar panel + battery.

Beneficiaries: Hill-state residents, disaster management cells.

PS-10 | AI-Powered Civic Issue Reporting App

Domain: Smart City | **Type:** 🖥️ Software | **Difficulty:** Beginner

Problem Statement: Potholes, garbage overflow, and broken streetlights go unreported or unresolved for weeks because there's no easy, trackable channel between citizens and municipal bodies.

Expected Solution:

- App where citizens photograph and geo-tag civic issues
- AI-based auto-categorization (pothole, garbage, streetlight, etc.) from the image
- Public status tracker (reported → assigned → resolved) for accountability
- Admin dashboard for municipal staff to prioritize by severity/area

Suggested Tech Stack: React Native/Flutter, a lightweight CNN for image classification, Node.js + PostgreSQL with PostGIS for geo-data.

Beneficiaries: Citizens, municipal corporations.

PS-11 | Adaptive Smart Streetlight System

Domain: Smart City | **Type:**  Hardware | **Difficulty:** Intermediate

Problem Statement: Municipal streetlights often run on fixed timers, wasting electricity on empty roads while poorly lighting high-traffic areas, inflating civic power bills.

Expected Solution:

- Motion/light sensor per streetlight adjusting brightness based on ambient light and presence of pedestrians/vehicles
- Central dashboard for the municipality to monitor faults and energy usage
- Automated fault detection (bulb failure alert) to reduce manual inspection

Suggested Tech Stack: LDR + PIR sensors, ESP32/Arduino, MQTT protocol, a simple monitoring dashboard.

Beneficiaries: Municipal corporations, city residents.

PS-12 | Rural Bus Tracking & Crowd Prediction App

Domain: Transportation | **Type:**  Software | **Difficulty:** Intermediate

Problem Statement: In smaller towns, state-run bus schedules are unreliable and commuters have no visibility into real-time location or expected crowding, leading to long unpredictable waits.

Expected Solution:

- Live GPS tracking of buses (using conductor's phone as a data source is acceptable for MVP)
- ETA prediction at each stop using historical + live data
- Crowd-level prediction/reporting (crowdsourced by passengers)
- Route and fare information in regional languages

Suggested Tech Stack: Google Maps/OpenStreetMap API, Firebase real-time database, Flutter.

Beneficiaries: Daily commuters, state transport corporations.

PS-13 | Two-Wheeler Accident Detection & Emergency Alert Device

Domain: Transportation | **Type:** ⚙️ Hardware | **Difficulty:** Advanced

Problem Statement: India has among the highest two-wheeler fatality rates partly due to delayed emergency response — accidents in remote stretches often go unnoticed for critical "golden hour" minutes.

Expected Solution:

- Accelerometer/gyroscope-based crash detection mounted on the vehicle
- Automatic SMS/call trigger to emergency contacts and nearest hospital with GPS location upon detecting a crash pattern
- Manual cancel button for false positives (e.g., pothole jolts)
- Optional helmet-integrated version with impact sensor

Suggested Tech Stack: MPU6050, SIM800L GSM module, NEO-6M GPS, Arduino/ESP32.

Beneficiaries: Two-wheeler riders, highway patrol/ambulance services.

PS-14 | Smart Waste Bin with Route Optimization

Domain: Waste Management | **Type:** 🔄 Hybrid | **Difficulty:** Intermediate

Problem Statement: Municipal garbage trucks follow fixed routes regardless of actual bin fill-levels, wasting fuel on empty bins while overflowing bins elsewhere go unattended.

Expected Solution:

- Ultrasonic fill-level sensor in bins transmitting status to a central server
- Route optimization algorithm generating the shortest collection path for only the bins that need pickup
- Dashboard for municipal waste management staff
- Citizen-facing map showing nearest available bins

Suggested Tech Stack: Ultrasonic sensor (HC-SR04), ESP32, Google OR-Tools/Dijkstra for routing, web dashboard.

Beneficiaries: Municipal sanitation departments, residents.

PS-15 | Water Quality & Contamination Sensor Network

Domain: Water Resources | **Type:** ⚙️ Hardware | **Difficulty:** Intermediate

Problem Statement: Groundwater and public water-supply contamination (fluoride, arsenic, turbidity, bacterial) is a major rural health hazard, often detected only after illness outbreaks.

Expected Solution:

- Low-cost multi-parameter water sensor node (pH, turbidity, TDS) deployed at supply points/wells
- Real-time alerting to local health/water authority when parameters cross safe limits (BIS standards)
- Historical data dashboard to identify contamination trends by region

Suggested Tech Stack: pH sensor, turbidity sensor, TDS sensor, ESP32, ThingSpeak/custom dashboard.

Beneficiaries: Rural households, Jal Jeevan Mission/PHED authorities.

PS-16 | Solar Microgrid Health Monitoring System

Domain: Energy | **Type:** ⚙️ Hardware | **Difficulty:** Advanced

Problem Statement: Off-grid solar microgrids serving rural hamlets frequently fail silently (battery degradation, panel faults) with no remote diagnostics, leading to prolonged outages before a technician visits.

Expected Solution:

- Sensor unit monitoring panel voltage/current, battery health, and load consumption
- Remote dashboard alerting operators to anomalies (e.g., sudden voltage drop, battery degradation)
- Predictive maintenance flag based on historical performance trends
- SMS alert to local operator when in areas without internet

Suggested Tech Stack: Voltage/current sensors (ACS712, voltage divider), ESP32, GSM module, cloud dashboard.

Beneficiaries: Rural electrification agencies, off-grid village communities.

PS-17 | Voice-Based Banking Assistant for Rural Users

Domain: Financial Inclusion | **Type:** 💻 Software | **Difficulty:** Intermediate

Problem Statement: Many rural and semi-literate users find app-based banking/UPI interfaces difficult to navigate, limiting adoption of digital financial services despite having bank accounts.

Expected Solution:

- Voice-command interface in regional languages for balance check, mini statement, and fund transfer (simulated banking API for hackathon)
- Simple confirmation flow using voice + PIN for security
- Offline/IVR fallback mode for users without smartphones

Suggested Tech Stack: Speech-to-text (Google/Whisper API), NLP intent recognition, a mock banking sandbox API, Android app.

Beneficiaries: Rural bank account holders, semi-literate users.

PS-18 | Farmer-to-Market Direct Selling Platform

Domain: AgriTech / MSME | **Type:** 🖥️ Software | **Difficulty:** Beginner

Problem Statement: Farmers often sell produce to middlemen at low prices due to lack of direct market access and real-time mandi price visibility, cutting significantly into their margins.

Expected Solution:

- Marketplace app connecting farmers directly to buyers/retailers
- Live mandi price integration (using government eNAM-style open data) for informed pricing
- Logistics/pickup request feature for transport coordination
- Rating system to build buyer-seller trust

Suggested Tech Stack: React/Flutter, Node.js + MongoDB, integration with public mandi price datasets.

Beneficiaries: Farmers, local retailers/wholesalers.

PS-19 | UPI/Phishing Fraud Detection Browser Extension

Domain: Cybersecurity | **Type:** 🖥️ Software | **Difficulty:** Advanced

Problem Statement: UPI and banking phishing scams are rising sharply in India, and most victims are non-technical users who cannot distinguish a fake payment link/site from a genuine one.

Expected Solution:

- Browser extension/app that scans URLs and UPI payment links against known phishing patterns and heuristics
- Real-time warning banner before a user submits credentials or completes a payment
- Community-reported blocklist that updates a shared database
- Simple risk-score explanation in plain language

Suggested Tech Stack: Chrome Extension API, a phishing-URL feature-based ML classifier (Python/scikit-learn), a backend blocklist API.

Beneficiaries: General banking/UPI users, especially first-time digital adopters.

PS-20 | Credential Leak & Dark Web Exposure Monitor

Domain: Cybersecurity | **Type:** 🖥️ Software | **Difficulty:** Advanced

Problem Statement: Individuals and small organizations rarely find out their credentials were leaked in a data breach until after fraud occurs, since monitoring tools are enterprise-priced and inaccessible to them.

Expected Solution:

- Tool where a user submits an email/domain to check against publicly known breach databases (e.g., Have I Been Pwned API)
- Periodic automated re-checks with alert notifications
- Educational guidance on next steps (password reset, 2FA) after a detected leak
- Clear scoping: use only legitimate breach-check APIs, never scrape or access dark web content directly

Suggested Tech Stack: HaveIBeenPwned API, Node.js/Python backend, email notification service (SendGrid).

Beneficiaries: Individuals, student clubs/small organizations managing shared accounts.

PS-21 | Smart Cane for the Visually Impaired

Domain: Assistive Tech | **Type:**  Hardware | **Difficulty:** Intermediate

Problem Statement: Traditional white canes only detect ground-level obstacles, leaving visually impaired users vulnerable to chest/head-level obstacles and unable to sense obstacles beyond arm's reach.

Expected Solution:

- Ultrasonic sensors mounted on a cane detecting obstacles at multiple heights
- Vibration/audio feedback indicating obstacle direction and distance
- Optional GPS module with a "guardian alert" feature for location sharing
- Lightweight, low-power, weatherproof design

Suggested Tech Stack: HC-SR04 ultrasonic sensors, vibration motor, Arduino Nano, optional NEO-6M GPS.

Beneficiaries: Visually impaired individuals.

PS-22 | Sign Language to Text/Speech Translator

Domain: Assistive Tech | **Type:**  Software | **Difficulty:** Advanced

Problem Statement: Communication barriers between hearing/speech-impaired individuals and others remain high due to limited public understanding of Indian Sign Language (ISL) and lack of accessible real-time translation tools.

Expected Solution:

- Camera-based gesture recognition model translating ISL gestures into text in real time
- Text-to-speech output for two-way communication
- Support for a core vocabulary set (extendable) relevant to daily/emergency use
- Simple UI usable on a basic smartphone camera

Suggested Tech Stack: MediaPipe/OpenCV for hand tracking, a CNN/LSTM gesture-classification model, TensorFlow Lite for mobile deployment.

Beneficiaries: Hearing/speech-impaired individuals and the people they interact with.

PS-23 | AR-Based Heritage & Tourism Guide

Domain: Tourism & Culture | **Type:**  Software | **Difficulty:** Intermediate

Problem Statement: Smaller heritage sites and local monuments (outside major tourist circuits) lack interpretive information, causing visitors to miss the historical and cultural significance of what they're seeing.

Expected Solution:

- App using AR to overlay historical information, images, or reconstructions when a user points their camera at a monument
- Audio-guided walking tours in multiple languages
- Offline mode for areas with poor connectivity
- Local artisan/guide marketplace integration for added tourism revenue

Suggested Tech Stack: AR Core/AR Kit or a lightweight web-AR library (AR.js), Unity (optional), a content CMS backend.

Beneficiaries: Tourists, local tourism boards, heritage conservation bodies.

PS-24 | Gram Panchayat Grievance Redressal Chatbot

Domain: Governance | **Type:**  Software | **Difficulty:** Beginner

Problem Statement: Villagers often don't know the correct department or process to raise grievances (water supply, roads, ration card issues), leading to unresolved complaints and repeated in-person visits to panchayat offices.

Expected Solution:

- WhatsApp/SMS-based chatbot allowing villagers to lodge complaints in simple regional-language text or voice notes
- Automatic categorization and routing to the correct department (simulated for hackathon)
- Status tracking via a unique complaint ID
- Analytics dashboard for panchayat officials to spot recurring issues

Suggested Tech Stack: WhatsApp Business API/Twilio, Dialogflow/Rasa, Node.js backend, PostgreSQL.

Beneficiaries: Rural citizens, Gram Panchayat administration.

PS-25 | Low-Cost Air Quality Monitoring IoT Device

Domain: Environment | **Type:**  Hardware | **Difficulty:** Intermediate

Problem Statement: Air quality monitoring stations are concentrated in major cities, leaving smaller towns without any local AQI data despite significant pollution from vehicles, construction, and stubble burning.

Expected Solution:

- Sensor node measuring PM2.5, PM10, CO2, and humidity
- Data transmission to a public dashboard showing real-time hyperlocal AQI
- Alert system for hazardous AQI spikes (e.g., near schools)
- Low-cost, replicable design for deployment across multiple points in a town

Suggested Tech Stack: PMS5003/SDS011 dust sensor, MQ-135 gas sensor, ESP32, ThingSpeak or a custom map dashboard.

Beneficiaries: Town residents, local pollution control boards, schools.

PS-26 | Crowdsourced Tree Plantation & Carbon Tracker

Domain: Environment | **Type:**  Software | **Difficulty:** Beginner

Problem Statement: Tree plantation drives (government and NGO-led) lack transparent, verifiable tracking of survival rates, so it's unclear how much real environmental impact these drives actually achieve.

Expected Solution:

- App where volunteers geo-tag and photograph planted saplings
- Periodic photo check-ins to verify survival over months (gamified with badges/leaderboard)
- Estimated carbon-offset calculation based on tree species and count
- Public map visualizing plantation density and survival rate by region

Suggested Tech Stack: React Native/Flutter, geo-tagging APIs, Firebase, a simple carbon-offset estimation formula.

Beneficiaries: NGOs, forest departments, environmentally conscious citizens.

PS-27 | Wearable Performance Tracker for Rural Sports Talent

Domain: Sports | **Type:**  Hybrid | **Difficulty:** Intermediate

Problem Statement: Talented young athletes in rural areas often go unnoticed by sports scouts and coaching programs due to lack of access to performance-tracking infrastructure available in urban academies.

Expected Solution:

- Wearable device tracking sprint speed, jump height, and stride metrics using accelerometer/gyroscope

- Companion app logging performance history and generating shareable athlete profiles
- Comparison against benchmark standards for various sports/age groups
- Option to submit profiles to district/state sports authorities or scouts

Suggested Tech Stack: MPU6050, ESP32/nRF microcontroller, BLE for data sync, Flutter app.

Beneficiaries: Rural student-athletes, sports academies, Khelo India-style talent scouting programs.

PS-28 | Blockchain-Based Land Record Verification

Domain: Blockchain / GovTech | **Type:**  Software | **Difficulty:** Advanced

Problem Statement: Land record disputes and fraud (duplicate sales, forged documents) are a major source of litigation in India due to fragmented, easily tampered paper-based or siloed digital records.

Expected Solution:

- Blockchain-based ledger recording land ownership transfers immutably
- QR-code/ID-based verification portal for buyers to check authentic ownership history
- Smart contract simulation for transfer approval workflow (registrar sign-off)
- Clear scoping note: hackathon build should be a working prototype/simulation, not a legally binding registry

Suggested Tech Stack: Hyperledger Fabric/Ethereum (testnet), Solidity smart contracts, React frontend, IPFS for document storage.

Beneficiaries: Landowners, property buyers, state revenue departments.

PS-29 | Deepfake & Regional-Language Fake News Detector

Domain: AI/ML | **Type:**  Software | **Difficulty:** Advanced

Problem Statement: Misinformation spreads rapidly via WhatsApp/social media in regional languages and increasingly through AI-generated deepfake videos/audio, and most fact-checking tools only cover English content.

Expected Solution:

- Text classifier trained to flag likely misinformation patterns in regional-language news/forwards
- Basic deepfake video/audio detection using artifact-analysis models (face-warping inconsistencies, audio spectral anomalies)
- Browser extension or WhatsApp-forward checker returning a credibility score with source links
- Clear disclaimer that outputs are probabilistic aids, not definitive verdicts

Suggested Tech Stack: BERT-based multilingual classifiers (IndicBERT), a pretrained deepfake-detection model (e.g., open-source FaceForensics-based models), Flask/FastAPI backend.

Beneficiaries: Social media users, fact-checking organizations, journalists.

PS-30 | Low-Cost Agri-Robot for Sowing & Weeding

Domain: Robotics | **Type:** 🌀 Hardware | **Difficulty:** Advanced

Problem Statement: Manual sowing and weeding are labor-intensive and increasingly costly for small farmers due to rising rural labor shortages, directly affecting planting timeliness and yield.

Expected Solution:

- Small wheeled/tracked robot capable of row-based seed dispensing at set intervals
- Basic weed-vs-crop visual differentiation (color/shape-based) to trigger a mechanical weeding arm or targeted micro-spray
- Remote/app-based manual override and path control
- Rugged, low-cost chassis suited for uneven farmland

Suggested Tech Stack: Raspberry Pi/Arduino, motor drivers, DC geared motors, basic OpenCV-based weed detection, RF/Bluetooth remote control.

Beneficiaries: Small and marginal farmers, agricultural labor cooperatives.

Suggested Website Structure

For your "Problem Statement" page, consider building this as filterable cards using the fields above:

- **Filters:** Domain (dropdown/chips), Type (Software/Hardware/Hybrid), Difficulty (Beginner/Intermediate/Advanced)
- **Card view:** ID, Title, Domain badge, Type icon, one-line problem summary
- **Detail view (on click):** Full problem statement, expected solution (as checklist), suggested tech stack, beneficiaries, and a "Select this problem statement" button for team registration
- **Optional additions:** "Claimed by Team X" tag once a team selects a PS (common in SIH-style portals to avoid duplicate selection), and a submission-count indicator per PS

If you're building this in React, I can also generate this content as a JSON file (array of objects) so it plugs directly into your components instead of you re-typing from markdown — just say the word.